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Materia:

PATRONES DE DISEÑO DE SOFTWARE

Unidad y tema:

Unidad #3 Proyecto decorador en equipo

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Grupo:

SC9A

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Horario:

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Objetivo

La implementación del patrón decorador para extender la funcionalidad de las hamburguesas dentro del sistema permitiendo agregar una nueva clase decoradora que registre y gestione las notas personalizadas del cliente cuando se realice cambios en los ingredientes. Buscando que sin la modificación de la clase principal hamburguesa se pueda incorporar información adicional como observación específica de pedido mendieta la decoración manteniendo así los principios de abierto y cerrado. Y el patrón Puente “Bridge” tiene como objetivo desacoplar la implementación de las clases, de manera que esas clases puedan ser modificadas independientemente sin necesidad de alterar el otro tipo de entrega de comida que se implemento en el programa.

Código realizado para Form1.cs

```
using System.Text;
using iText.Kernel.Pdf;
using iText.Layout;
using iText.Layout.Element;
using System.IO;
using System.Drawing.Imaging;

namespace Carlsjr_Patrones
{
    public partial class Form1 : Form
    {
        public Form1()
        {
            InitializeComponent();
            CargarDatosIniciales();
            pictureBoxPreview.Visible = false;
            pictureBox6.Visible = false;
            pictureBox7.Visible = false;
        }

        private void CargarDatosIniciales()
        {
            cmbHamburguesa.Items.Add("Famous Star");
            cmbHamburguesa.Items.Add("Western Bacon");
            cmbHamburguesa.Items.Add("Super Star");
            cmbHamburguesa.SelectedIndex = 0;

            cmbPapas.Items.Add("Sin papas");
```

```
    cmbPapas.Items.Add("Papas a la francesa");
    cmbPapas.Items.Add("Papas Crisscut");
    cmbPapas.Items.Add("Papas Curly");
    cmbPapas.SelectedIndex = 0;

    cmbBebida.Items.Add("Sin bebida");
    cmbBebida.Items.Add("Refresco chico");
    cmbBebida.Items.Add("Refresco mediano");
    cmbBebida.Items.Add("Refresco grande");
    cmbBebida.SelectedIndex = 0;

    cmbEntrega.Items.Add("En restaurante");
    cmbEntrega.Items.Add("Para llevar");
    cmbEntrega.Items.Add("Drive Thru");
    cmbEntrega.SelectedIndex = 0;
}

private Hamburguesa CrearHamburguesaBase()
{
    switch (cmbHamburguesa.Text)
    {
        case "Famous Star":
            return new FamousStar();
        case "Western Bacon":
            return new WesternBacon();
        case "Super Star":
            return new SuperStar();
        default:
            return new FamousStar();
    }
}

private Papas CrearPapas()
{
    switch (cmbPapas.Text)
    {
        case "Papas a la francesa":
            return new PapasFrancesa();
        case "Papas Crisscut":
            return new PapasCrisscut();
        case "Papas Curly":
            return new PapasCurly();
        default:
            return new SinPapas();
    }
}
```

```
private Bebida CrearBebida()
{
    switch (cmbBebida.Text)
    {
        case "Refresco chico":
            return new RefrescoChico();
        case "Refresco mediano":
            return new RefrescoMediano();
        case "Refresco grande":
            return new RefrescoGrande();
        default:
            return new SinBebida();
    }
}

private ITipoEntrega CrearTipoEntrega()
{
    switch (cmbEntrega.Text)
    {
        case "En restaurante":
            return new EnRestaurante();
        case "Para llevar":
            return new ParaLlevar();
        case "Drive Thru":
            return new DriveThru();
        default:
            return new EnRestaurante();
    }
}

private void btnGenerarPedido_Click(object sender, EventArgs e)
{
    Hamburguesa hamburguesa = CrearHamburguesaBase();

    if (chkQueso.Checked)
        hamburguesa = new ExtraQueso(hamburguesa);

    if (chkTocino.Checked)
        hamburguesa = new ExtraTocino(hamburguesa);

    if (chkJalapenos.Checked)
        hamburguesa = new Jalapenos(hamburguesa);

    if (chkArosCebolla.Checked)
        hamburguesa = new ArosDeCebolla(hamburguesa);
```

```
        if (chkGuacamole.Checked)
            hamburguesa = new Guacamole(hamburguesa);

        if (chkSinCebolla.Checked)
            hamburguesa = new SinCebolla(hamburguesa);

        if (chkSinTomate.Checked)
            hamburguesa = new SinTomate(hamburguesa);

        if (chkSinPepinillos.Checked)
            hamburguesa = new SinPepinillos(hamburguesa);

        Papas papas = CrearPapas();
        Bebida bebida = CrearBebida();
        ITipoEntrega entrega = CrearTipoEntrega();

        Pedido pedido = new PedidoCarlsJr(hamburguesa, papas, bebida,
entrega);

        StringBuilder sb = new StringBuilder();
        sb.AppendLine(pedido.MostrarResumen());
        sb.AppendLine($"TOTAL: ${pedido.CalcularTotal():0.00}");

        txtResumen.Text = sb.ToString();
        lblTotal.Text = $"Total: ${pedido.CalcularTotal():0.00}";
    }

    private void btnLimpiar_Click(object sender, EventArgs e)
    {
        cmbHamburguesa.SelectedIndex = 0;
        cmbPapas.SelectedIndex = 0;
        cmbBebida.SelectedIndex = 0;
        cmbEntrega.SelectedIndex = 0;

        chkQueso.Checked = false;
        chkTocino.Checked = false;
        chkJalapenos.Checked = false;
        chkArosCebolla.Checked = false;
        chkGuacamole.Checked = false;

        chkSinCebolla.Checked = false;
        chkSinTomate.Checked = false;
        chkSinPepinillos.Checked = false;

        txtResumen.Clear();
        lblTotal.Text = "Total: $0.00";
    }
```

```
        private void Form1_Load(object sender, EventArgs e)
        {

        }

        private void cmbHamburguesa_DrawItem(object sender, DrawItemEventArgs
e)
        {
            if (e.Index < 0) return; // Skip if no item is selected

            ComboBox comboBox = sender as ComboBox;
            Brush textBrush;
            Brush backgroundBrush;

            // Check if the current item is selected
            if ((e.State & DrawItemState.Selected) == DrawItemState.Selected)
            {
                // Customize colors for selected item (e.g., blue background, yellow text)
                backgroundBrush = new SolidBrush(Color.FromArgb(254, 198, 39));
                textBrush = new SolidBrush(Color.Black);
            }
            else
            {
                // Customize colors for non-selected items (e.g., white background, black
text)
                backgroundBrush = new SolidBrush(Color.FromArgb(95, 95, 95));
                textBrush = new SolidBrush(Color.FromArgb(254, 198, 39));
            }

            // Draw the background of the item
            e.Graphics.FillRectangle(backgroundBrush, e.Bounds);

            // Draw the item text
            e.Graphics.DrawString(comboBox.Items[e.Index].ToString(), e.Font,
textBrush, e.Bounds, StringFormat.GenericDefault);

            // Draw the focus rectangle if the item has focus
            e.DrawFocusRectangle();

            // Clean up the brushes
            textBrush.Dispose();
            backgroundBrush.Dispose();
        }

        private void txtResumen_TextChanged(object sender, EventArgs e)
        {
```

```
}

private void groupBoxExtras_Enter(object sender, EventArgs e)
{
    // vacío, no hace nada
}

private void chkSinTomate_CheckedChanged(object sender, EventArgs e)
{

}

private void groupBoxQuitar_Enter(object sender, EventArgs e)
{

}
//Extras
private bool extrasExpandido = false;
private bool quitarExpandido = false;



private void label1_Click(object sender, EventArgs e)
{


}

private void label2_Click(object sender, EventArgs e)
{

}



private void pictureBox4_Click(object sender, EventArgs e)
{
    if (!extrasExpandido)
    {
        // EXPANDIR
        groupBoxExtras.Size = new Size(250, 193);
        groupBoxQuitar.Location = new Point(608, 120);
        pictureBox5.BringToFront();


        label1.Text = "-";
```

```
            extrasExpandido = true;
        }
        else
        {
            // CONTRAER
            groupBoxExtras.Size = new Size(250, 32);
            groupBoxQuitar.Location = new Point(608, 120);
            pictureBox5.BringToFront();


            label1.Text = "+";

            extrasExpandido = false;
        }
    }

    private void pictureBox5_Click(object sender, EventArgs e)
    {
        if (!quitarExpandido)
        {
            // EXPANDIR
            groupBoxQuitar.Size = new Size(250, 129);
            label2.Text = "-";

            quitarExpandido = true;
        }
        else
        {
            // CONTRAER
            groupBoxQuitar.Size = new Size(250, 32);
            label2.Text = "+";

            quitarExpandido = false;
        }
    }

    private void btnImprimir_Click(object sender, EventArgs e)
    {
        SaveFileDialog saveFileDialog = new SaveFileDialog();
        saveFileDialog.Filter = "PDF (*.pdf)|*.pdf";
        saveFileDialog.FileName = "Pedido_CarlsJr.pdf";

        if (saveFileDialog.ShowDialog() == DialogResult.OK)
        {
            string path = saveFileDialog.FileName;
```

```
            using (PdfWriter writer = new PdfWriter(path))
            using (PdfDocument pdf = new PdfDocument(writer))
            using (Document document = new Document(pdf))
            {
                // Título
                document.Add(new Paragraph("===== PEDIDO CARL'S JR =====")
                    //.SetBold()
                    .SetFontSize(16));

                // Contenido (tu resumen)
                document.Add(new Paragraph(txtResumen.Text));

                // Total
                document.Add(new Paragraph(lblTotal.Text));


                document.Close();
            }

            MessageBox.Show("Recibo creado correctamente ✅");
        }
    }

    // Designer-referenced event handler stubs (prevent designer/CodeDom
parsing errors)
    private void cmbHamburguesa_SelectedIndexChanged(object sender,
EventArgs e)
    {
        switch (cmbHamburguesa.Text)
        {
            case "Famous Star":
                pictureBoxPreview.Visible = true;

                pictureBoxPreview.Image = Properties.Resources.Famous;
                break;

            case "Western Bacon":
                pictureBoxPreview.Visible = true;

                pictureBoxPreview.Image = Properties.Resources.Western;
                break;

            case "Super Star":
                pictureBoxPreview.Visible = true;

                pictureBoxPreview.Image = Properties.Resources.Super;
                break;
```

```
        }
    }

    private void cmbBebida_SelectedIndexChanged(object sender, EventArgs e)
    {
        if (cmbBebida.Text == "Sin bebida")
        {
            pictureBox7.Visible = false;
        }
        else
        {
            pictureBox7.Image = Properties.Resources.Bebida; // una sola imagen
            pictureBox7.Visible = true;
        }
    }

    private void pictureBoxPreview_Click(object sender, EventArgs e)
    {
        // Optional: show a larger preview or no-op. Kept to satisfy designer wiring.
    }

    private void cmbPapas_SelectedIndexChanged(object sender, EventArgs e)
    {
            switch (cmbPapas.Text)
{
    case "Sin papas":
                pictureBox6.Visible = false;
                break;

    case "Papas a la francesa":
                pictureBox6.Visible = true;

                pictureBox6.Image = Properties.Resources.French;
        break;

    case "Papas Crisscut":
                pictureBox6.Visible = true;

                pictureBox6.Image = Properties.Resources.Crisscut;
        break;

    case "Papas Curly":
                pictureBox6.Visible = true;

                pictureBox6.Image = Properties.Resources.Curly;
        break;
}
```

```
        }
    }
}
```

Clase realizada para "PedidoCarlsJr.cs"

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    public class PedidoCarlsJr : Pedido
    {
        public PedidoCarlsJr(Hamburguesa hamburguesa, Papas papas, Bebida bebida, ITipoEntrega
tipoEntrega)
            : base(hamburguesa, papas, bebida, tipoEntrega)
        {
        }

        public override string MostrarResumen()
        {
            return
                "===== PEDIDO CARL'S JR =====" + Environment.NewLine +
                $"Hamburguesa: {hamburguesa.GetDescripcion()}" + Environment.NewLine +
                $"Papas: {papas.GetDescripcion()}" + Environment.NewLine +
                $"Bebida: {bebida.GetDescripcion()}" + Environment.NewLine +
                $"Tipo de entrega: {tipoEntrega.ProcesarEntrega()}" + Environment.NewLine +
Environment.NewLine +
                $"Hamburguesa: ${hamburguesa.GetCosto():0.00}" + Environment.NewLine +
                $"Papas: ${papas.GetCosto():0.00}" + Environment.NewLine +
                $"Bebida: ${bebida.GetCosto():0.00}" + Environment.NewLine +
                $"Cargo entrega/servicio: ${tipoEntrega.CostoEntrega():0.00}";
        }

        public override decimal CalcularTotal()
        {
            return hamburguesa.GetCosto()
                + papas.GetCosto()
                + bebida.GetCosto()
                + tipoEntrega.CostoEntrega();
        }
    }
}
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    public class PedidoCarlsJr : Pedido
    {
        public PedidoCarlsJr(Hamburguesa hamburguesa, Papas papas, Bebida bebida, ITipoEntrega
tipoEntrega)
            : base(hamburguesa, papas, bebida, tipoEntrega)
        {
```

```
        }

        public override string MostrarResumen()
        {
            return
                "===== PEDIDO CARL'S JR =====" + Environment.NewLine +
                $"Hamburguesa: {hamburguesa.GetDescripcion()}" + Environment.NewLine +
                $"Papas: {papas.GetDescripcion()}" + Environment.NewLine +
                $"Bebida: {bebida.GetDescripcion()}" + Environment.NewLine +
                $"Tipo de entrega: {tipoEntrega.ProcesarEntrega()}" + Environment.NewLine +
Environment.NewLine +
                $"Hamburguesa: ${hamburguesa.GetCosto():0.00}" + Environment.NewLine +
                $"Papas: ${papas.GetCosto():0.00}" + Environment.NewLine +
                $"Bebida: ${bebida.GetCosto():0.00}" + Environment.NewLine +
                $"Cargo entrega/servicio: ${tipoEntrega.CostoEntrega():0.00}";
        }

        public override decimal CalcularTotal()
        {
            return hamburguesa.GetCosto()
                + papas.GetCosto()
                + bebida.GetCosto()
                + tipoEntrega.CostoEntrega();
        }
    }
}
```

Clase Realizada para "Pedido.cs"

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    public abstract class Pedido
    {
        protected ITipoEntrega tipoEntrega;
        protected Hamburguesa hamburguesa;
        protected Papas papas;
        protected Bebida bebida;

        public Pedido(Hamburguesa hamburguesa, Papas papas, Bebida bebida, ITipoEntrega tipoEntrega)
        {
            this.hamburguesa = hamburguesa;
            this.papas = papas;
            this.bebida = bebida;
            this.tipoEntrega = tipoEntrega;
        }

        public abstract string MostrarResumen();
        public abstract decimal CalcularTotal();
    }

}
```

Clases realizadas para el Patrón “Decorador”

Clase Hamburguesa.cs

Carlsjr_Patrones | Carlsjr_Patrones.Hamburguesa

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       20 references
9      public abstract class Hamburguesa
10     {
           20 references
11         public abstract string GetDescripcion();
           21 references
12         public abstract decimal GetCosto();
13     }
```

Clase FamousStar.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.FamousStar

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       2 references
9      public class FamousStar : Hamburguesa
10     {
           10 references
11         public override string GetDescripcion() => "Famous Star";
           11 references
12         public override decimal GetCosto() => 110.00m;
13     }
14 }
15
```

Clase SuperStar.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.SuperStar

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    1 reference
    public class SuperStar : Hamburguesa
    {
        10 references
        public override string GetDescripcion() => "Super Star";
        11 references
        public override decimal GetCosto() => 145.00m;
    }
}
```

Clase WesternBacon.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.WesternBacon

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    1 reference
    public class WesternBacon : Hamburguesa
    {
        10 references
        public override string GetDescripcion() => "Western Bacon";
        11 references
        public override decimal GetCosto() => 135.00m;
    }
}
```

Clase para HamburguesaDecorator.cs

```
Carlsjr_Patrones                                   Carlsjr_Patrones.HamburguesaDecorator
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        17 references
9       public abstract class HamburguesaDecorator : Hamburguesa
10      {
11          protected Hamburguesa hamburguesa;
12
            8 references
13          public HamburguesaDecorator(Hamburguesa hamburguesa)
14          {
15              this.hamburguesa = hamburguesa;
16          }
17      }
18  }
```

Clase para ArosDeCebolla.cs

```
Form1.cs [Design]                                                    ArosDeCebolla.cs
Carlsjr_Patrones        Carlsjr_Patrones.ArosDeCebolla        ArosDeCebolla(Hamburguesa hamburguesa)
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        2 references
9       public class ArosDeCebolla : HamburguesaDecorator
10      {
            1 reference
11          public ArosDeCebolla(Hamburguesa hamburguesa) : base(hamburguesa) { }
            10 references
12          public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Aros de cebolla";
            11 references
13          public override decimal GetCosto() => hamburguesa.GetCosto() + 20.00m;
14      }
15  }
```

Clase para ExtraQueso.cs

```
Form1.cs [Design]                                                    ExtraQueso
Carlsjr_Patrones        Carlsjr_Patrones.ExtraQueso        ExtraQueso(Hamburguesa hamburguesa)
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        2 references
9       public class ExtraQueso : HamburguesaDecorator
10      {
            1 reference
11          public ExtraQueso(Hamburguesa hamburguesa) : base(hamburguesa) { }
            10 references
12          public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Extra queso";
            11 references
13          public override decimal GetCosto() => hamburguesa.GetCosto() + 18.00m;
14      }
15  }
```

Clase para ExtraTocino.cs

```
Form1.cs [Design]
Carlsjr_Patrones        Carlsjr_Patrones.ExtraTocino        ExtraTocino(Hamburguesa hamb
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        2 references
9       public class ExtraTocino : HamburguesaDecorator
10      {
            1 reference
11          public ExtraTocino(Hamburguesa hamburguesa) : base(hamburguesa) { }
            10 references
12          public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Tocino";
            11 references
13          public override decimal GetCosto() => hamburguesa.GetCosto() + 22.00m;
14      }
15  }
```

Clase para Guacamole.cs

```
Carlsjr_Patrones        Carlsjr_Patrones.Guacamole        Guacamole(Hamburguesa hamburgu
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        2 references
9       public class Guacamole : HamburguesaDecorator
10      {
            1 reference
11          public Guacamole(Hamburguesa hamburguesa) : base(hamburguesa) { }
            10 references
12          public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Guacamole";
            11 references
13          public override decimal GetCosto() => hamburguesa.GetCosto() + 25.00m;
14      }
15  }
```

Clase para Jalapenos.cs

```
Carlsjr_Patrones        Carlsjr_Patrones.Jalapenos        Jalapenos(Hamburguesa hamburgue
1  using System;
2  ng System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       2 references
9      public class Jalapenos : HamburguesaDecorator
10     {
           1 reference
11         public Jalapenos(Hamburguesa hamburguesa) : base(hamburguesa) { }
           10 references
12         public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Jalapeños";
           11 references
13         public override decimal GetCosto() => hamburguesa.GetCosto() + 12.00m;
14     }
15 }
16
```

Clase para SinCebolla.cs

```
Form1.cs [Design]                                                                SinCebol
Carlsjr_Patrones        Carlsjr_Patrones.SinCebolla        SinCebolla(Hamburguesa hamburguesa)
1  using System;
2  ng System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       2 references
9      public class SinCebolla : HamburguesaDecorator
10     {
           1 reference
11         public SinCebolla(Hamburguesa hamburguesa) : base(hamburguesa) { }
           10 references
12         public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Sin cebolla";
           11 references
13         public override decimal GetCosto() => hamburguesa.GetCosto();
14     }
15 }
16
```

Clase para SinPepinillos.cs

```
Carlsjr_Patrones        Carlsjr_Patrones.SinPepinillos        SinPepinillos(Hamburguesa hamburguesa)
1  using System;
2  ng System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       2 references
9      public class SinPepinillos : HamburguesaDecorator
10     {
           1 reference
11         public SinPepinillos(Hamburguesa hamburguesa) : base(hamburguesa) { }
           10 references
12         public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Sin pepinillos";
           11 references
13         public override decimal GetCosto() => hamburguesa.GetCosto();
14     }
15 }
```

Clase para SinTomate.cs

```
Carlsjr_Patrones                    Carlsjr_Patrones.SinTomate          SinTomate(Hamburguesa hamburguesa
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        2 references
9       public class SinTomate : HamburguesaDecorator
10      {
            1 reference
11          public SinTomate(Hamburguesa hamburguesa) : base(hamburguesa) { }
            10 references
12          public override string GetDescripcion() => hamburguesa.GetDescripcion() + ", Sin tomate";
            11 references
13          public override decimal GetCosto() => hamburguesa.GetCosto();
14      }
15  }
```

Clase para Bebidas.cs

```
Form1.cs [Design]
Carlsjr_Patrones                    Carlsjr_Patrones.Bebida
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        9 references
9       public abstract class Bebida
10      {
            5 references
11          public abstract string GetDescripcion();
            6 references
12          public abstract decimal GetCosto();
13      }
14  }
```

Clase para RefrescoChico.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.RefrescoChico

```
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        1 reference
9       public class RefrescoChico : Bebida
10      {
            2 references
11          public override string GetDescripcion() => "Refresco chico";
            3 references
12          public override decimal GetCosto() => 20.00m;
13      }
14  }
```

Clase para RefrescoMediano.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.RefrescoMediano

```
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        1 reference
9       public class RefrescoMediano : Bebida
10      {
            2 references
11          public override string GetDescripcion() => "Refresco mediano";
            3 references
12          public override decimal GetCosto() => 28.00m;
13      }
14  }
```

Clase para RefrescoGrande.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.RefrescoGrande

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       1 reference
9      public class RefrescoGrande : Bebida
10     {
           2 references
11         public override string GetDescripcion() => "Refresco grande";
           3 references
12         public override decimal GetCosto() => 35.00m;
13     }
14 }
```

Clase para SinBebida.cs

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.SinBebida

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       1 reference
9      public class SinBebida : Bebida
10     {
           2 references
11         public override string GetDescripcion() => "Sin bebida";
           3 references
12         public override decimal GetCosto() => 0.00m;
13     }
14 }
```

Clase para [Papas.cs]

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.Papas

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       9 references
9      public abstract class Papas
10     {
           5 references
11         public abstract string GetDescripcion();
           6 references
12         public abstract decimal GetCosto();
13     }
14 }
```

Clases para [PapasCrissCut.cs]

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.PapasCrisscut

```
1  using System;
2  using System.Collections.Generic;
3  using System.Linq;
4  using System.Text;
5  using System.Threading.Tasks;
6
7  namespace Carlsjr_Patrones
8  {
       1 reference
9      public class PapasCrisscut : Papas
10     {
           2 references
11         public override string GetDescripcion() => "Papas Crisscut";
           3 references
12         public override decimal GetCosto() => 42.00m;
13     }
14 }
```

Clase para PapasCurly.cs

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    1 reference
    public class PapasCurly : Papas
    {
        2 references
        public override string GetDescripcion() => "Papas Curly";
        3 references
        public override decimal GetCosto() => 40.00m;
    }

}
```

Clase para PapasFrancesa.cs

```
Carlsjr_Patrones                    Carlsjr_Patrones.PapasFrancesa
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    1 reference
    public class PapasFrancesa : Papas
    {
        2 references
        public override string GetDescripcion() => "Papas a la francesa";
        3 references
        public override decimal GetCosto() => 35.00m;
    }
}
```

Clase para SinPapas.cs

```
Form1.cs [Design]
Carlsjr_Patrones                    Carlsjr_Patrones.SinPapas
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        1 reference
9       public class SinPapas : Papas
10      {
            2 references
11          public override string GetDescripcion() => "Sin papas";
            3 references
12          public override decimal GetCosto() => 0.00m;
13      }
14  }
```

Clases realizadas para el Patrón “Bridge”

Clase Para ITipoEntrega.cs

```
Form1.cs [Design]
Carlsjr_Patrones                    Carl
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        8 references
9       public interface ITipoEntrega
10      {
            4 references
11          string ProcesarEntrega();
            5 references
12          decimal CostoEntrega();
13      }
14  }
```

Clase para EnRestaurante.cs

```
Carlsjr_Patrones                    Carlsjr_Patrones.EnRestaurante
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        2 references
9       public class EnRestaurante : ITipoEntrega
10      {
            2 references
11          public string ProcesarEntrega() => "Consumo en restaurante";
            3 references
12          public decimal CostoEntrega() => 0m;
13      }
14  }
15
```

Clase para DriveThru.cs

```
Form1.cs [Design]
Carlsjr_Patrones                    Carlsjr_Patrones.DriveThru
1   using System;
2   using System.Collections.Generic;
3   using System.Linq;
4   using System.Text;
5   using System.Threading.Tasks;
6
7   namespace Carlsjr_Patrones
8   {
        1 reference
9       public class DriveThru : ITipoEntrega
10      {
            2 references
11          public string ProcesarEntrega() => "Pedido por Drive Thru";
            3 references
12          public decimal CostoEntrega() => 8m;
13      }
14  }
```

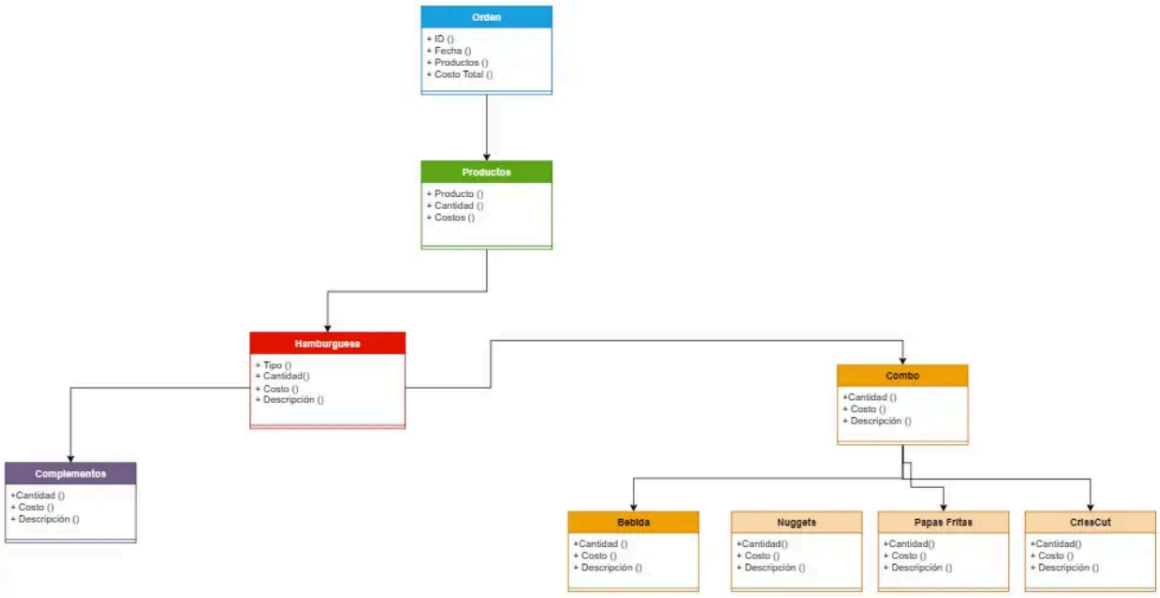
Clase para [ParaLlevar.cs]

Form1.cs [Design]

Carlsjr_Patrones | Carlsjr_Patrones.ParaLlevar

```
using System;
using System.Collections.Generic;
using System.Linq;
using System.Text;
using System.Threading.Tasks;

namespace Carlsjr_Patrones
{
    1 reference
    public class ParaLlevar : ITipoEntrega
    {
        2 references
        public string ProcesarEntrega() => "Pedido para llevar";
        3 references
        public decimal CostoEntrega() => 5m;
    }
}
```

Diagrama UML

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